What Makes People Happy?

An Analysis of the World Happiness Report 2023

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1. Introduction

- The **World Happiness Report (WHR)** has measured subjective well-being in 150+ countries since 2012. Respondents rate life satisfaction on the **Cantril ladder**: 0 = worst, 10 = best possible life.
- Key factors studied: economic output, social support, health, and freedom.

2. Theoretical Background

Research Question

Which socioeconomic factors correlate most strongly with subjective well-being?

Hypotheses

1. Countries with **higher GDP per capita** report higher happiness.
2. **Stronger social support** is positively associated with happiness.

3. Method

- **Dataset:** World Happiness Report 2023 (Helliwell et al., 2023)
- **Sample:** 137 countries (WHR 2023 uses 2020 to 2022 survey averages)
- **Instrument:** Gallup World Poll, Cantril ladder (0 to 10)
- **Analysis:** Pearson correlation (r) and significance test (p) in R

4. Results

- **H1:** GDP per capita correlates strongly with happiness (r **= 0.78,** p **<0.001**; Fig. 1).
- **H2:** Social support is an even stronger predictor (r **= 0.83,** p **<0.001**; Fig. 2).
- Fig. 3: **GDP** and **Social Support** contribute most to the score. Bar size reflects the factor's numerical scale, not r.
- Fig. 4: Ranking of all 6 WHR factors by correlation $|r|$. **Social Support** and **GDP** lead; Generosity is essentially uncorrelated with happiness.

5. Discussion & Conclusion

- Both hypotheses confirmed. Social support outperforms GDP as a predictor by correlation (r = 0.83 vs. r = 0.78; Fig. 4).
- *Why is social support the better predictor?* A higher r means points cluster more tightly around the trend line. Countries with high social support *reliably* show high happiness, so we can predict a country's happiness more accurately from its social support than from its GDP.
- *Why does GDP contribute more "absolute points" (Fig. 3)?* Fig. 3 shows how many score points each factor *adds on average*, not how well it predicts. Logged GDP takes values from 7 to 11, social support only 0 to 1. The same variation therefore translates into more numeric points for GDP, purely because its scale is wider. This is a **scale effect**, not a stronger relationship. Correlation r is scale free, so Fig. 4 is the fair comparison.
- *Limitation:* Correlation $\neq$ causation; self-reports may be culturally biased.
- Conclusion: **Economic security** and **social cohesion** are the key drivers of well-being.

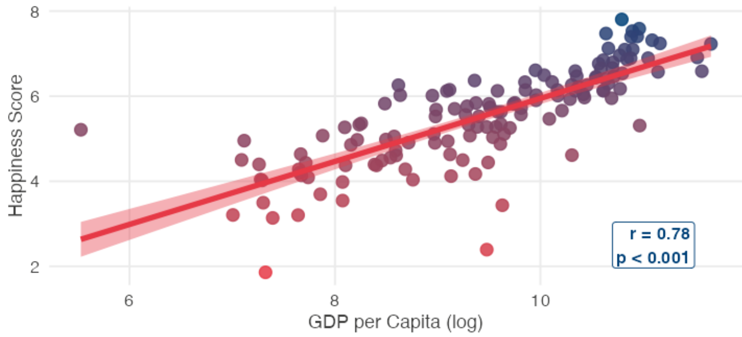

Figure 1: GDP & Happiness (H1). 137 countries; r **= 0.78**, p **<0.001**. Point colour: red = low happiness, navy = high. Red line = OLS regression.

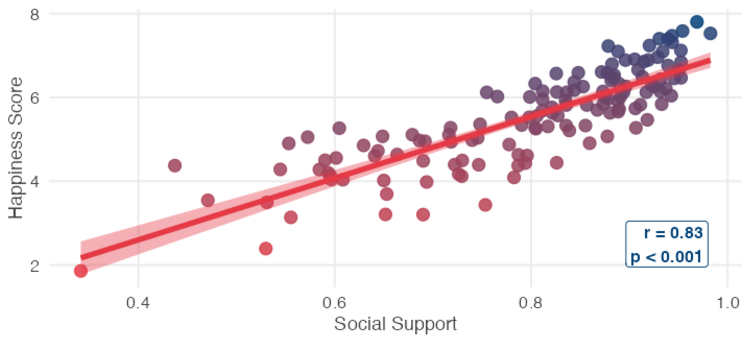

Figure 2: Social Support & Happiness (H2). 137 countries; r **= 0.83**, p **<0.001**. Same style as Fig. 1. Points hug the regression line more tightly than in Fig. 1, which is what a higher correlation looks like.

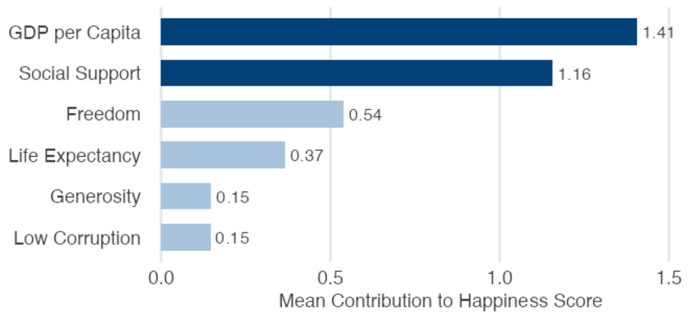

Figure 3: Factor Contributions. Mean absolute contribution of all 6 WHR factors to the happiness score. **Navy** = factors tested here. GDP leads due to its larger numerical scale, not stronger correlation (r).

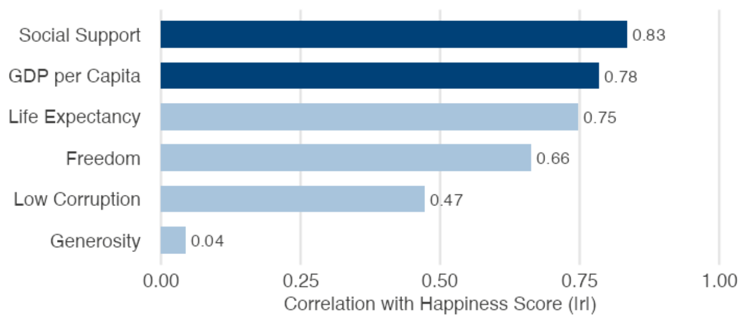

Figure 4: Factor Correlations. Pearson $|r|$ of each factor with the happiness score. Unlike Fig. 3, this removes the scale effect: **Social Support** now leads GDP, confirming it is the stronger predictor.

References

Helliwell, J. F. et al. (2023). *World Happiness Report 2023.* Sustainable Development Solutions Network (SDSN).

Gallup Inc. (2023). *Gallup World Poll Methodology.* gallup.com